NOTE: The document identifier and heading have been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

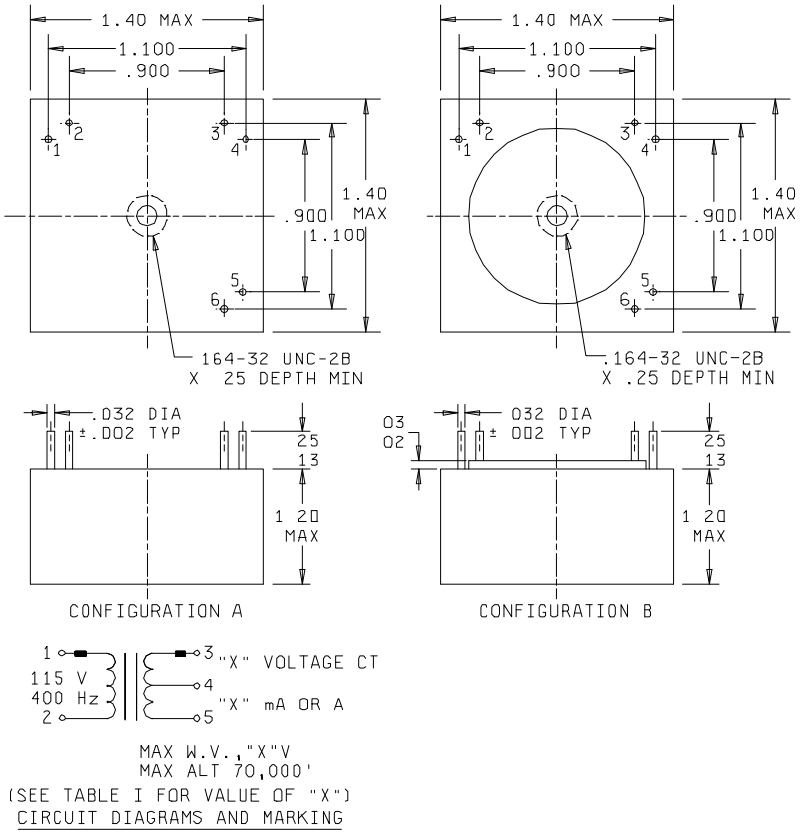
MIL-PRF-27/46B
7 February 1984
SUPERSEDING
MIL-T-27/46A
20 October 1976

PERFORMANCE SPECIFICATION SHEET

TRANSFORMERS, POWER, 25 VOLT AMPERES, 400 HZ, TF5S03ZZ796 TO TF5S03ZZ839, INCLUSIVE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the transformers described herein shall consist of this specification and the latest issue of MIL-T-27.



INCHES	MM
.002	0.05
.02	0.5
.03	0.8
.032	0.81
.13	3.3
.164	4.17
.25	6.4
.900	22.86
1.00	25.4
1.10	27.9
1.20	30.5
1.40	35.6

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Tolerance on terminal position dimensions is ±.005 (0.13 mm).
4. Marking shall be on the side of the case.
5. A white dot shall be marked on the side opposite the terminals, over terminal 1, in addition to the terminal marking shown.
6. Secondary electrical values shall be marked as specified in table I, as applicable.

FIGURE 1. Dimensions and configuration.

(B) denotes changes

FSC 5950

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T03:17Z
Check the source to verify that this is the current version before use.

REQUIREMENTS: (When numbers in parentheses i.e. (1-2) are used, they indicate the winding and the extreme terminals of the winding.)

Electrical ratings See table I.

Primary voltage (1-2) 115 volts ±10 percent, 400 Hz ±5 percent.

Design and construction

Dimensions and configuration See figure 1

Weight. 4.4 ounces, maximum.

Duty cycle: Continuous.

Case. Plastic encapsulated.

Material Plastic.

Altitude: 70,000 feet, maximum.

Operating temperature range -55°C to +130°C maximum.

Terminals: Printed circuit pins - tinned brass.

Terminal strength. Method 211 of MIL-STD-202, test condition A, 2.0 pounds.

Dielectric withstanding voltage

At sea level. 2.8 times the maximum working voltage specified in table I.

At reduced barometric pressure 1.25 times the maximum working voltage specified in table I.

Electrical characteristics.

No load. With 115 volts, 400 Hz across (1-2) Current in (1-2) shall not exceed 25 mA rms.

Rated load With 115 volts, 400 Hz across (1-2) and a resistive load with current as specified in table I in (3-5), the voltage across (3-5) shall be as specified in table I.

Efficiency 88 percent minimum at rated load.

Polarity Additive with terminals 2 and 3 connected.

Regulation $\frac{\text{Voltage (no load)} - \text{Voltage (rated load)}}{\text{Voltage (rated load)}} \times 100$ shall not exceed 7 percent

Phase shift 1.0 electrical degree between primary and secondary voltage at rated load, maximum.

Temperature rise: 45°C, maximum.

Vibration. MIL-STD-202, method 204, test condition D.

Marking location See figure 1.

Ⓑ QUALITY ASSURANCE PROVISIONS:

Extent of qualification

Qualification testing and approval to M27/37-44 A or B, M27/42-44 A or B, and M27/46-44 A or B shall be sufficient to grant qualification approval to MIL-T-27/37 through MIL-T-27/47 inclusive, all parts.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T03:17Z
Check the source to verify that this is the current version before use.

Qualification testing and approval to M27/46-44 shall be sufficient to grant qualification approval to M27/46-01 through M27/46-44.

Part number example:

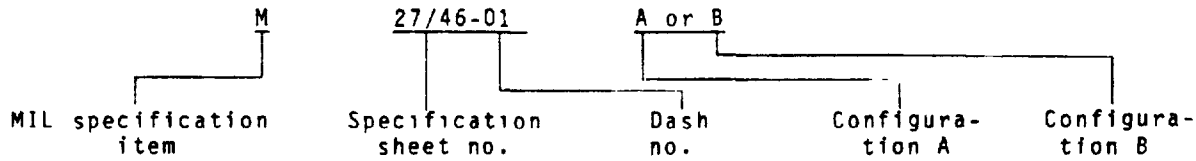


TABLE I. Electrical ratings.

Dash no.	Type designation	Secondary voltage (3-5) (V rms) ±3%	Secondary current (3-5) (amps rms)	Working voltage (V rms max)
01 A or B	796	6	4.167	325
02 "	797	6.3	3.968	"
03 "	798	7	3.571	"
04 "	799	7.5	3.333	"
05 "	800	8	3.125	"
06 "	801	9	2.778	"
07 "	802	10	2.500	"
08 "	803	11	2.273	"
09 "	804	12	2.083	"
10 "	805	12.6	1.984	"
11 "	806	13	1.923	"
12 "	807	15	1.667	"
13 "	808	16	1.563	"
14 "	809	18	1.389	"
15 "	810	20	1.250	"
16 "	811	22	1.136	"
17 "	812	24	1.042	"
18 "	813	26	0.962	"
19 "	814	28	0.893	"
20 "	815	30	0.833	"
21 "	816	33	0.758	"
22 "	817	36	0.694	"
23 "	818	39	0.641	"
24 "	819	43	0.581	"
25 "	820	47	0.532	"
26 "	821	51	0.490	"
27 "	822	56	0.446	"
28 "	823	62	0.403	"
29 "	824	68	0.367	"
30 "	825	75	0.333	"
31 "	826	82	0.305	"
32 "	827	91	0.275	"
33 "	828	100	0.250	"
34 "	829	110	0.227	"
35 "	830	115	0.218	"
36 "	831	120	0.208	"
37 "	832	130	0.192	"
38 "	833	150	0.166	"
39 "	834	180	0.139	"
40 "	835	200	0.125	440
41 "	836	220	0.113	440
42 "	837	240	0.104	440
43 "	838	270	0.093	580
44 "	839	300	0 083	580

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T03:17Z
Check the source to verify that this is the current version before use.

Custodians.
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - OS
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - ME
Navy - MC, AS
Air Force - 19

Agent:
DLA - ES

Preparing activity
Army - ER

(Project 5950-0623-12)

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T03:17Z
Check the source to verify that this is the current version before use.